Project Documentation: Predictive Maintenance & Anomaly Detection for IoT Fleets

Table of Contents

Overview

This project demonstrates an **end-to-end MLOps pipeline** for real-time anomaly detection in IoT sensor data. It simulates a fleet of vehicles sending telemetry (temperature, vibration, RPM) to Kafka, processes the stream with Apache Spark, stores raw data in HDFS and real-time anomalies in MongoDB, aggregates data nightly into PostgreSQL, trains an Isolation Forest model with MLflow, orchestrates retraining with Kubeflow, and serves predictions via a FastAPI endpoint.

The entire system is containerised with Docker Compose and designed for **Any** company specialising in AI, IoT, and smart city solutions.

Business Value

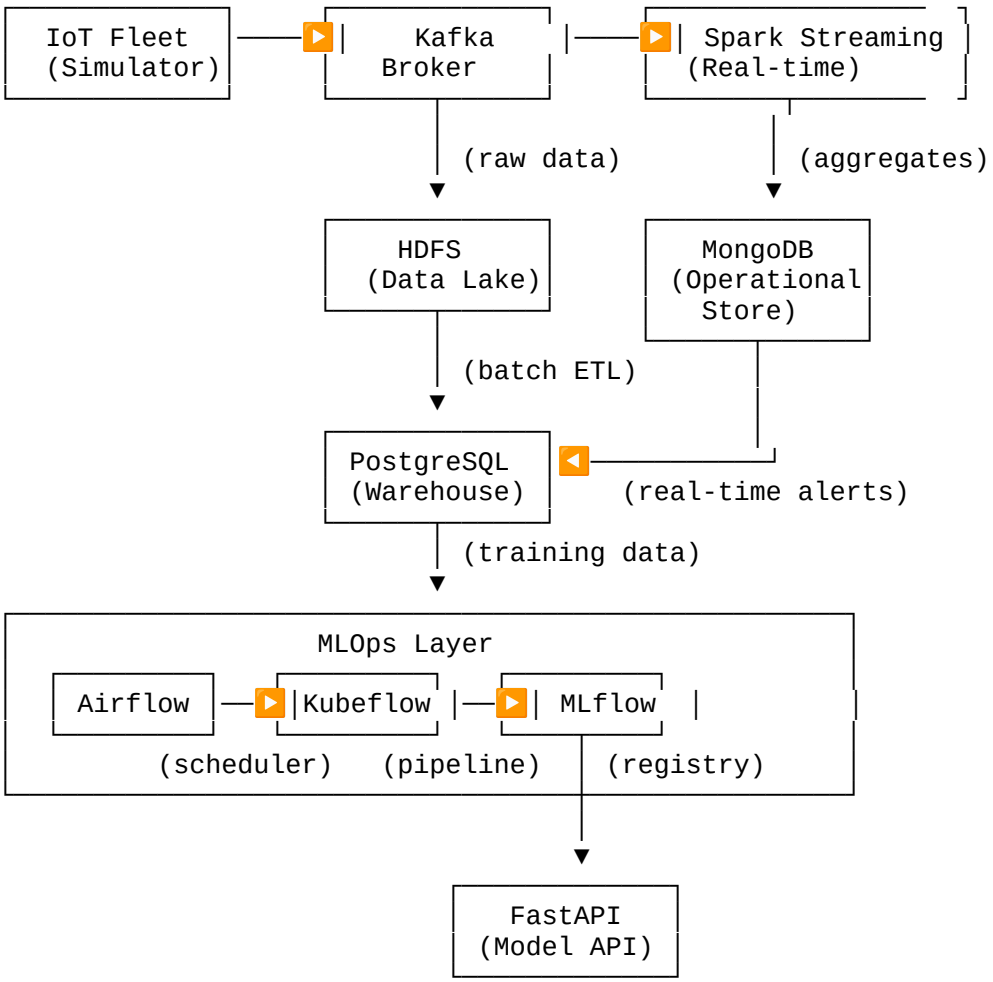
- **Real-time anomaly detection** – immediate alerts for critical failures (e.g., overheating, excessive vibration).

- **Scalable data ingestion** – Kafka and Spark handle millions of events per second.
- **Data lake + warehouse** – HDFS for long-term storage, PostgreSQL for business analytics.
- **MLOps governance** – MLflow tracks experiments, Kubeflow automates retraining.
- **Production-ready serving** – FastAPI/KServe provides low-latency predictions.
- **Full observability** – Prometheus + Grafana monitor pipeline health.

This architecture is directly applicable to Orion Valley's smart city, logistics, and industrial IoT use cases.

Architecture Diagram

text



Data flows from left to right; MLOps layer orchestrates model retraining on a schedule.

Technology Stack

Layer	Technology	Version
Stream Ingestion	Apache Kafka + Zookeeper	3.4.0 / latest

Layer	Technology	Version
Stream Processing	Apache Spark (Structured Streaming)	3.3.2
Data Lake	Hadoop HDFS (Parquet)	3.2.1
Operational DB	MongoDB	6.0
Warehouse	PostgreSQL	14
Workflow Orchestration	Apache Airflow	2.6.0
ML Orchestration	Kubeflow Pipelines	1.7
Experiment Tracking & Registry	MLflow	2.3.2
Model Serving	FastAPI	0.95.0
Monitoring	Prometheus + Grafana	latest
Containerisation	Docker, Docker Compose	20.10+
CI/CD	GitHub Actions	-
Language	Python	3.9

Project Structure

text

```
orion-valley-predictive-maintenance/
├── .github/workflows/
│   └── ci.yml                    # CI pipeline (unit tests only)
├── data/
│   └── generator/
│       └── kafka_producer.py     # Simulated IoT data source
├── processing/
│   ├── streaming/
│   │   └── streaming_anomaly_job.py   # Spark streaming job
│   └── batch/
│       └── hdfs_to_postgres.py        # Batch ETL (Airflow)
├── ml_pipeline/
│   ├── train_isolation_forest.py      # Training with MLflow
│   └── kubeflow/
│       └── pipeline.py                # Kubeflow pipeline definition
├── orchestration/
│   └── dags/
│       └── batch_etl_dag.py           # Airflow DAG
├── serving/
│   └── api/
│       └── fastapi_app.py             # Model serving API
├── monitoring/
│   └── prometheus.yml                 # Prometheus scrape config
├── tests/
│   ├── unit/                          # Unit tests
│   └── integration/                   # Integration tests (local only)
├── scripts/
│   └── setup_test_data.sh             # Seed PostgreSQL
├── docker-compose.yml                 # All services
├── hadoop.env                         # HDFS config
├── requirements.txt
├── pytest.ini
├── Makefile
└── README.md
```

Prerequisites

- **Docker** (20.10+) and **Docker Compose** (v2)
- **Python 3.9+** with `pip`
- **Java 11** (for Spark local mode)
- **Git**
- **8 GB RAM** recommended (for all containers)

Installation & Setup

1. Clone the repository

bash

```
git clone
https://github.com/MohsenMostafa1/Supply-Chain-Risk-Intelligence/tree/main.git
cd orion-valley-predictive-maintenance
```

2. Create a Python virtual environment

bash

```
python -m venv venv
source venv/bin/activate   # Linux/macOS
# or .\venv\Scripts\activate (Windows)
```

3. Install Python dependencies

bash

```
pip install -r requirements.txt
```

4. Start all services with Docker Compose

bash

```
docker-compose up -d
```

This starts:

- Zookeeper & Kafka
- Hadoop HDFS (Namenode + Datanode)
- Spark Master & Worker
- PostgreSQL
- MongoDB
- MLflow tracking server
- Airflow webserver & scheduler
- Prometheus

- Grafana

Note: Wait about 60 seconds for all services to fully initialise.

5. Verify services are running

bash

```
docker-compose ps
```

All services should show `Up` or `running`.

6. Seed test data (optional)

bash

```
bash scripts/setup_test_data.sh
```

Running the Pipeline Step by Step

Step 1: Generate simulated IoT data

bash

```
make produce
# or: python data/generator/kafka_producer.py
```

This script sends JSON messages to Kafka topic `iot-sensor-data` at 1 message per second (rotating among 100 vehicles).

Step 2: Start the Spark streaming job

In a new terminal (with the virtual environment activated):

bash

```
make stream
# or: spark-submit --packages org.apache.spark:spark-sql-kafka-0-10_2.12:3.3.0 \
#      processing/streaming/streaming_anomaly_job.py
```

The streaming job:

- Reads from Kafka
- Computes 10-second sliding-window averages per vehicle
- Flags anomalies when average vibration > 3.0
- Writes raw data to HDFS (`hdfs://localhost:9000/user/raw_iot/`)
- Writes anomalies to MongoDB (`orion.anomalies` collection)

Step 3: Train a model and log to MLflow

bash

```
make train
# or: python ml_pipeline/train_isolation_forest.py
```

- Loads historical data from PostgreSQL (table `features`)
- Trains an Isolation Forest model
- Logs parameters, metrics, and model to MLflow
- Registers the model as `OrionAnomalyDetector`

Visit http://localhost:5000 to see the MLflow UI.

Step 4: Serve the model via API

bash

```
make serve
# or: uvicorn serving.api.fastapi_app:app --reload --port 8000
```

Test the API:

bash

```
curl -X POST http://localhost:8000/predict \
  -H "Content-Type: application/json" \
  -d '{"temperature": 110, "vibration": 4.2, "rpm": 5800}'
```

Response: `{"anomaly": true, "score": -1}`

Step 5: Trigger batch ETL (Airflow)

- Open Airflow UI at http://localhost:8081 (user: `airflow`, password: `airflow`)
- Find DAG `orion_batch_etl_and_retraining`
- Trigger it manually (or wait for nightly 2 AM schedule)

This DAG:

1. Runs `hdfs_to_postgres.py` – aggregates daily data from HDFS to PostgreSQL
2. Triggers the Kubeflow pipeline (if deployed) to retrain the model

Step 6: Monitor with Grafana

- Open Grafana at http://localhost:3000 (user: `admin`, password: `admin`)
- Add Prometheus data source: `http://prometheus:9090`
- Import a dashboard (e.g., Kafka consumer lag, Spark metrics)

Testing

Unit Tests

Unit tests cover:

- Data generator logic
- Anomaly scoring function
- (No external dependencies – they run in CI)

Run locally:

bash

```
pytest tests/unit/ -v
```

Integration Tests (Local)

Integration tests verify that Kafka, Spark, and MongoDB work together. They require all Docker services to be running.

bash

```
docker-compose up -d   # ensure services are up
pytest tests/integration/ -v
```

> **Note**: Integration tests are **not** executed in GitHub Actions due to Kafka startup reliability issues. They are fully functional locally.

CI/CD Pipeline

The GitHub Actions workflow (`.github/workflows/ci.yml`) runs on every push and pull request:

- Python 3.9 setup
- Dependency installation
- Linting with `flake8`
- Unit tests with coverage reporting
- Coverage upload to Codecov (optional)

Integration tests are **disabled** in CI but documented for local execution.

Troubleshooting

Kafka doesn’t start in Docker Compose

- Increase memory limit for Docker (minimum 4 GB).
- Use Bitnami images (already in `docker-compose.yml`).
- Wait longer – Kafka can take up to 60 seconds.

Spark cannot connect to Kafka

- Ensure Kafka is up: `docker-compose ps`
- Check Kafka logs: `docker-compose logs kafka`
- Verify topic exists: `docker-compose exec kafka kafka-topics --list --bootstrap-server localhost:9092`

MLflow fails to connect to PostgreSQL

- Ensure PostgreSQL is running: `docker-compose ps postgres`
- Check the connection string in `docker-compose.yml` for MLflow (uses `postgres` hostname, not `localhost`).

HDFS commands fail

- Namenode may need extra time: `docker-compose logs hadoop-namenode`
- Use `hdfs dfs -ls /` inside the namenode container.

FastAPI model not found

- Train the model first (`make train`)
- Ensure the model file path in `fastapi_app.py` is correct (or load from MLflow registry).

Extending the Project

Add a new data source (e.g., weather API)

- Create a new Kafka producer that pushes weather data
- Join it with vehicle data in Spark

Replace Isolation Forest with a deep learning model (LSTM)

- Modify `train_isolation_forest.py` to use TensorFlow/PyTorch
- Log the Keras model with MLflow using `mlflow.tensorflow.log_model()`

Deploy to Kubernetes (production)

- Use `kompose` to convert `docker-compose.yml` to K8s manifests
- Replace localhost references with Kubernetes service names
- Use KServe instead of FastAPI for serverless inference

Add real-time dashboard (Streamlit)

- Read from MongoDB `orion.anomalies` every second
- Display live anomaly alerts and vehicle status

License